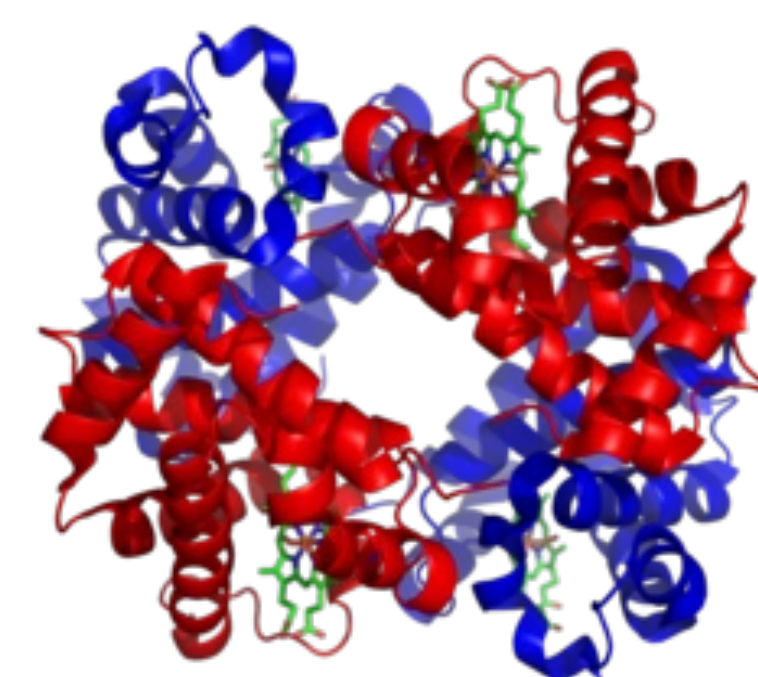
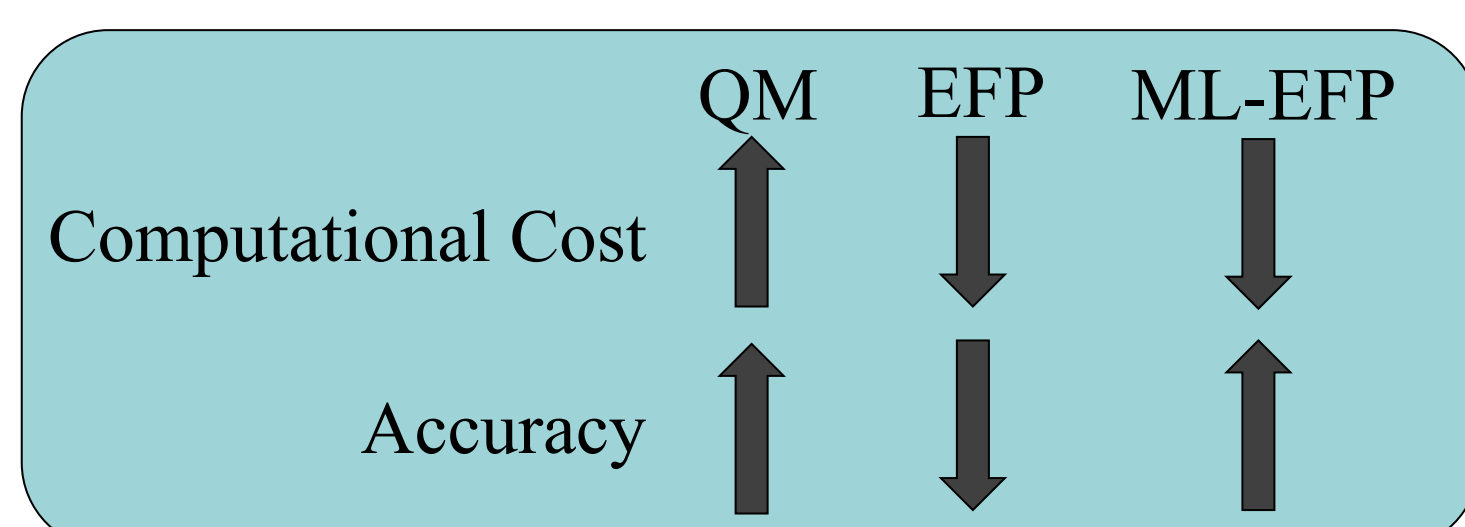


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Evaluation of Effective Fragmentation Potentials for Solvated Metal Ion–Amino Acid Systems

Introduction & Background

- Existing benchmarks, such as the S22 dataset, primarily focus on organic molecular interactions.
- Metal ion systems are important because they play critical roles in:
 - Enzyme catalysis and regulation
 - Protein folding and structural stability
 - Metalloprotein function and biological signaling
- Quantum mechanical (QM) methods provide accurate interaction energies but become computationally expensive for large molecular systems.
- Effective Fragment Potential (EFP) models provide QM-like interaction accuracy while significantly reducing computational cost.



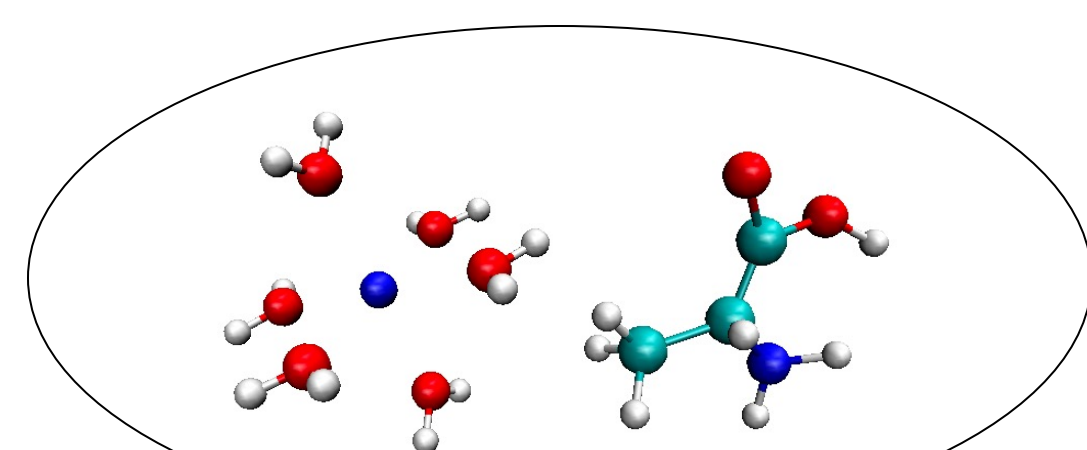
The Journal of Physical Chemistry A 2001, 105 (2), 293–307.
Wikipedia. *Hemoglobin*. Wikipedia 2026, 1, 1.

Objectives:

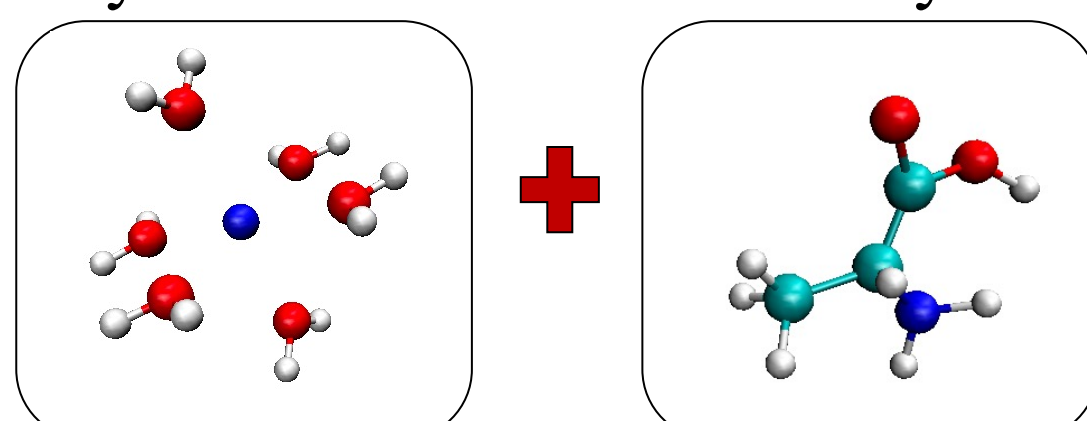
- Develop EFP models for metal ion–amino acid interaction systems.
- Establish a dataset of metal ion interactions to expand beyond existing organic-focused computational databases.
- Evaluate the accuracy of EFP models by comparing EFP interaction energies with QM reference calculations.

What is an EFP?

- Quantum mechanical method used to model noncovalent interactions between molecular fragments.
- Represents molecules as polarizable fragments rather than explicitly calculating every electron.
- EFP2 enables efficient modeling of metal ion–amino acid interactions by treating molecular environments as interacting fragments.



Hydrated Sodium–Alanine System



- Quantum Mechanic Model**
 - Molecules treated as a single system in which all electrons are computed explicitly across every atom simultaneously

- EFP Model**
 - Each molecule is a pre-parameterized fragment; interactions are computed through energy terms rather than explicit electrons

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Dampening Functions

$$E^{EFP} = E^{Coul} + E^{Pol} + E^{Disp} + E^{ExRep} + E^{CT}$$

The total EFP interaction energy is the sum of the electrostatic, polarization, dispersion, exchange repulsion, and charge transfer energy components.

Two types of f_n^{disp} :

Overlap-based

$$f_n^S(i, j) = 1 - S_{ij}^2 \sum_{n=0}^{6 \text{ or } 7 \text{ or } 8} \frac{-2 \ln |S_{ij}|^{n/2}}{n!}$$

Tang-Toennies

$$f_n^{TT}(i, j) = 1 - \left(\sum_{n=0}^{6 \text{ or } 7 \text{ or } 8} \frac{(bR_{ij})^n}{n!} \right) e^{-bR_{ij}}$$

The EFP electrostatic, polarization, and dispersion energies use damping functions from a summation of pairwise interactions at localized sites

$$E^{Disp} = E(R^{-6}) + E(R^{-7}) + E(R^{-8}) = \sum_{n=6,7,8} E(R^{-n})$$

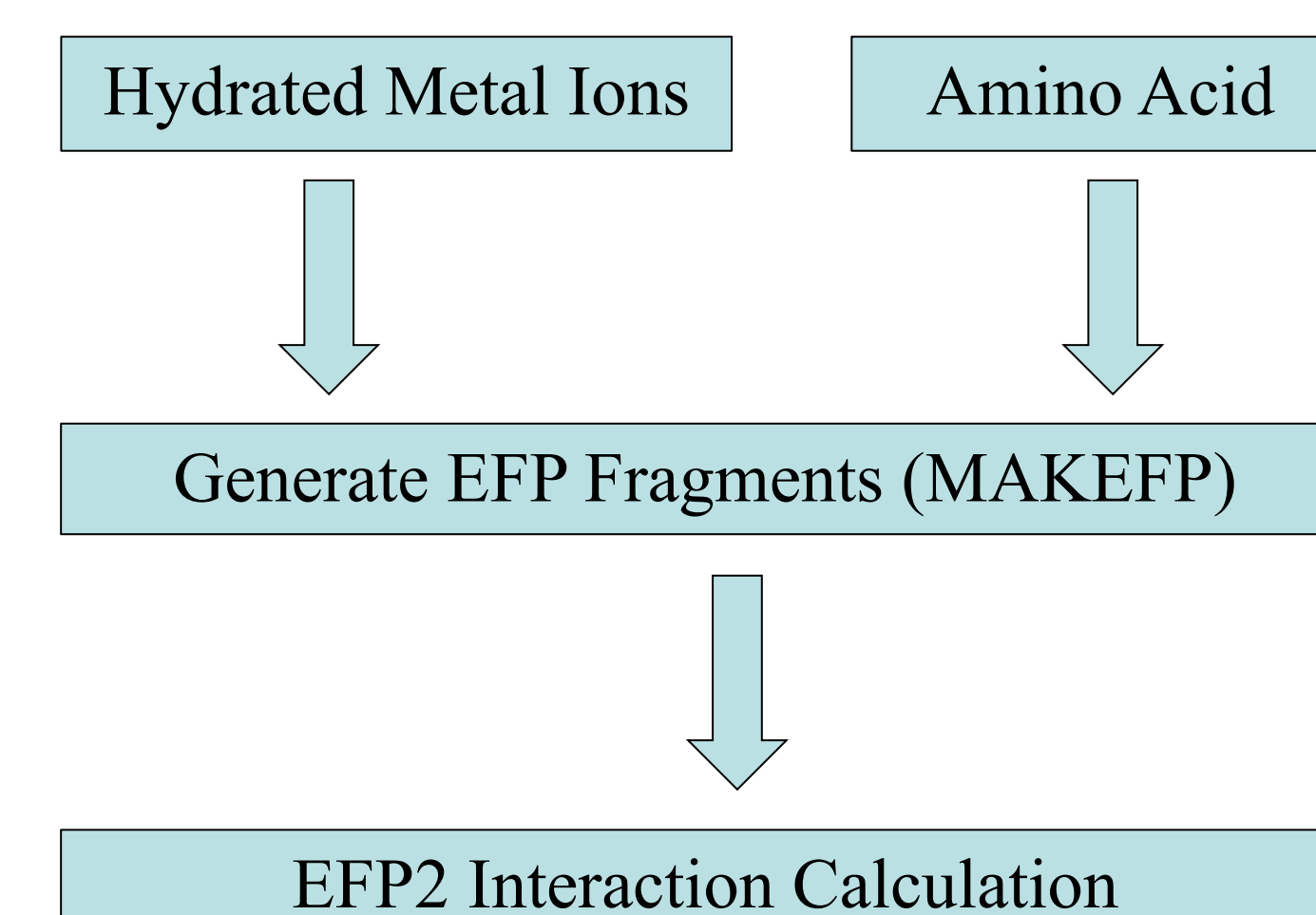
The EFP dispersion energy combines the R^{-6} , R^{-7} , and R^{-8} dispersion terms with overlap based and distance based damping functions

$$E(R^{-n}) = f_n^{disp} * \text{analytic expression}$$

Each term is multiplied by a damping function to account for short range interactions, where undamped potentials are inaccurate

Methods

- Constructed hydrated metal ions and amino acid structures
- Generated EFP fragments for each solvated metal ion and amino acid using MAKEFP
- Combined metal ion and amino acid fragments to perform EFP2 calculations

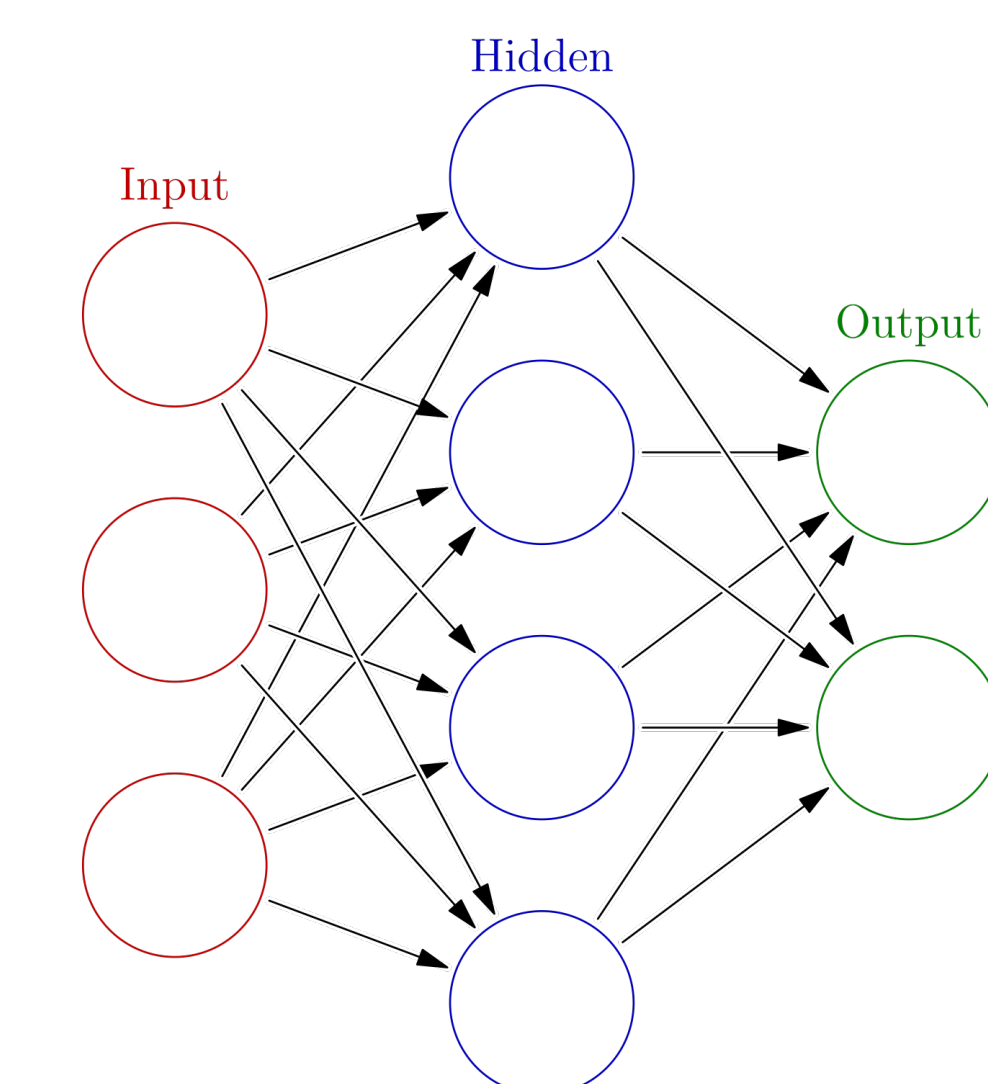


Metal Ion Systems Studied		
Metal Ion	Charge	Biological role
Na ⁺	+1	- Ion transport
K ⁺	+1	- Ion balance - Cellular signaling
Mg ²⁺	+2	- Enzyme activity - Cofactor function

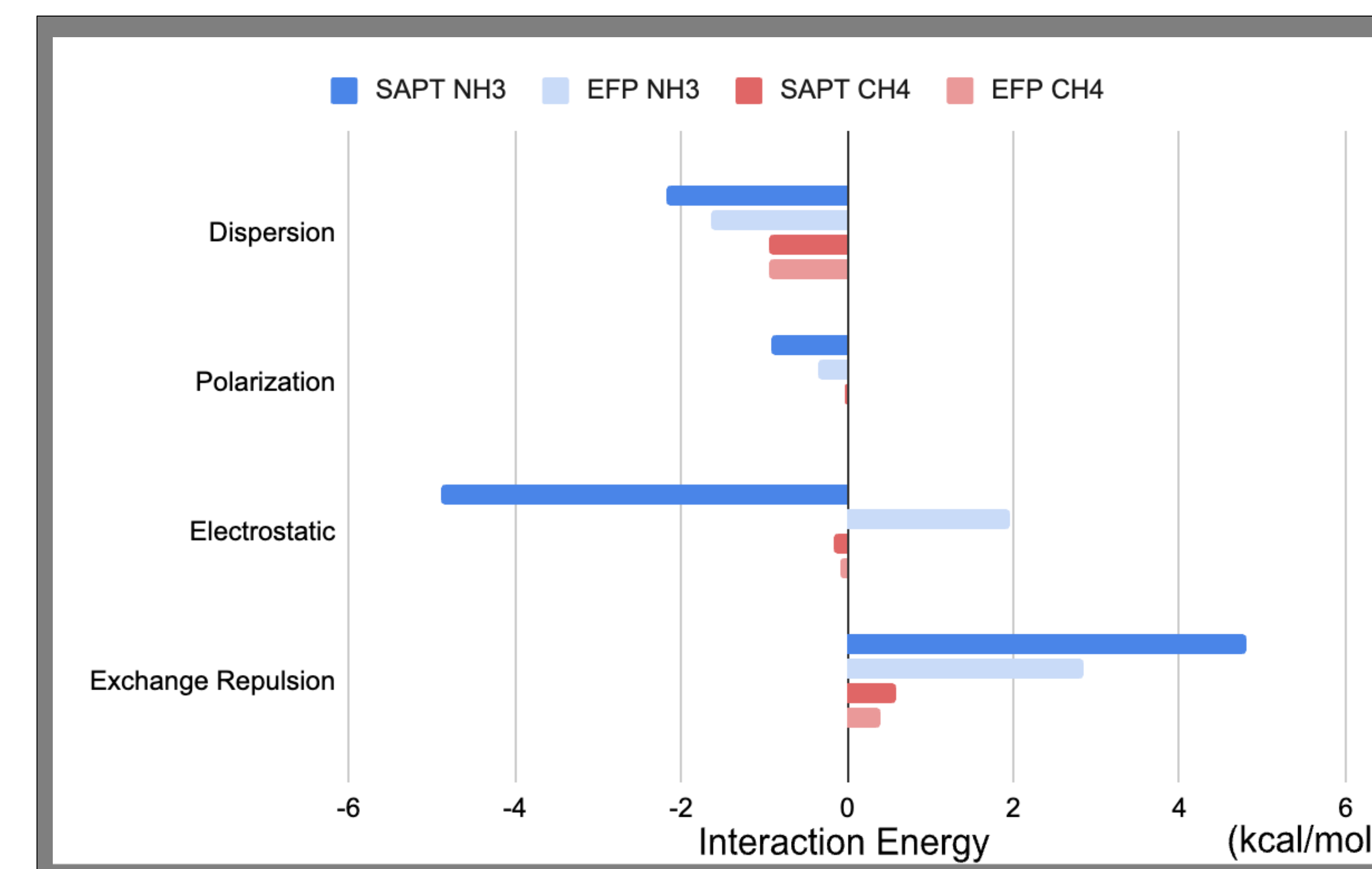
- Expand metal ion–amino acid EFP2 benchmarks beyond current organic molecule datasets to better represent biologically relevant interactions.
- Integrate optimized EFP models into larger biomolecular simulations to improve computational studies of metal binding environments.

Journal of Molecular Modeling 2013, 19 (11), 4661–4667.

- Apply ML-guided damping function optimization to improve EFP accuracy and enhance modeling of metal protein interactions.
- Uses calculated EFP dispersion energies as inputs and SAPT dispersion energies as the target (ground truth) for training damping function models (e.g., linear, exponential).



Data



Metal Protein Example						
Example	Electrostatic Energy	Polarization Energy	Dispersion Energy	Repulsion Energy	Charge Transfer	Final EFP Energy
Na ⁺ _Alanine	-1.1132	-0.5350	-0.3337	0.1149	0.0000	-1.8674

Conclusions

- Successfully calculated EFP total and interaction energies for metal–peptide systems and compared the results to SAPT reference values.
- Generated a library of EFP fragments for hydrated metal ions and representative amino acids, providing benchmark systems for machine learning development.
- Established a library of metal–peptide dimers prepared for the development and evaluation of new ML-EFP damping functions.
- Created a benchmark dataset and computational workflow to support the next stage of machine learning enhanced EFP model development.